

MATH 3312, SPRING 2026: SOLUTIONS TO HOMEWORK 2

(1) Determine if the following statements are true or false. Give a line or two in justification.

(a) Every finite integral domain is a field.

Solution. True. We saw this in Problem 6 of Homework 1.

(b) There exists a finite field of order 24.

Solution. False. Such a field would have to have elements of additive order both 2 and 3 (why?), but a finite field has a unique prime characteristic.

(c) There exists a finite field of order 8.

Solution. True. The polynomial $x^3 + x + 1 \in \mathbb{F}_2[x]$ is irreducible (why?) of degree 3, and so the quotient ring $\mathbb{F}_2[x]/(x^3 + x + 1)$ is a field of order 8 (why?).

(d) There are no ring homomorphisms¹

$$\mathbb{Z}[x]/(x^2 + 1) \rightarrow \mathbb{F}_7.$$

Solution. True. Giving such a ring homomorphism amounts to giving an element $a \in \mathbb{F}_7$ such that $a^2 = -1$. But you can check that no such element exists.

(e) There exists a finite field k with $k^\times \simeq \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$.

Solution. False. $k^\times$ is always cyclic for any finite field.

(2) Give examples of the following:

(a) A finite field of order 49.

Solution. As 1(d) indicates, the polynomial $x^2 + 1 \in \mathbb{F}_7[x]$ is irreducible (why?), and so the quotient ring $\mathbb{F}_7[x]/(x^2 + 1)$ is an example.

(b) A generator for $\mathbb{F}_{11}^\times$.

Solution. This is a group of order 10, so we are looking for an element that doesn't have order 2 or 5. Start with the smallest possibility: $2^5 = 32 = -1 \in \mathbb{F}_{11}$. This shows that 2 has order 10 and so it does the job. Other generators are obtained by taking that powers of 2 that are relatively prime to 10 (why?): $2^3 = 8, 2^7 = 7, 2^9 = 6$.

(c) A finite field with no elements of (multiplicative) order 3.

Solution. We know that $\mathbb{F}_p^\times$ is a cyclic group of order $p - 1$, and so it is enough to find a prime p such that $3 \nmid p - 1$. There are any number of options. Clearly $p = 2$ does the job, but also $p = 5, 11, 17$, etc.

(3) Let k be a finite field of characteristic p (so that it contains $\mathbb{F}_p$).

(a) Show that, for all elements $\alpha, \beta \in k$, we have

$$(\alpha + \beta)^p = \alpha^p + \beta^p.$$

¹The problem as stated had the word *non-trivial* in it, but any ring homomorphism has to be non-trivial, since it has to take 1 to 1.

Solution. This requires the *binomial formula*:

$$(\alpha + \beta)^p = \sum_{i=0}^p \binom{p}{i} \alpha^i \beta^{p-i}.$$

What happens now is that, for $i \neq 0, p$,

$$\binom{p}{i} = \frac{p!}{i!(p-i)!}$$

still has a factor of p in the numerator, and so is $0 \pmod{p}$. This leaves us with the terms corresponding to $i = 0, p$, which gives us the desired identity.

- (b) Show that $1 \in k$ is the only zero for $x^p - 1 \in k[x]$.

Solution. This is because if α is a zero for $x^p - 1$, then we would get

$$0 = \alpha^p - 1 = \alpha^p + (-1)^p = (\alpha - 1)^p \Rightarrow \alpha = 1.$$

Here, we have of course used the identity in (a). Note that the identity $-1 = (-1)^p$ holds in $\mathbb{F}_p$ for all primes p , including $p = 2$!

- (c) Show that the function $\text{Fr} : \alpha \mapsto \alpha^p$ is a ring *automorphism* from k to itself, and that $\text{Fr}(a) = a$ for all $a \in \mathbb{F}_p$.

Solution. We have to first show Fr respects addition and multiplication: The first follows from (a), and the second is the easy identity $(\alpha\beta)^p = \alpha^p\beta^p$. This tells us that Fr is a ring homomorphism, and to know that it is an *automorphism*, we need to know that Fr is bijective. Since we are dealing with finite sets, it is sufficient to know that it is *injective*. For this, note that if $a \in \ker \text{Fr}$, then we have $a^p = 0$, which of course implies that $a = 0$ (since we are in a field). Alternatively, *any* ring homomorphism out of a field is injective: see Observation 4 on Lecture 4.

- (4) Show that the following are equivalent for a field k of order p^d (where p is prime):

- (a) $k^\times$ contains an element of order n .
 (b) $p^d \equiv 1 \pmod{n}$.

Solution. Since $k^\times$ is cyclic of order $p^d - 1$, we are simply saying that a cyclic group of order $p^d - 1$ contains an element of order n if and only if $n \mid p^d - 1$. The only if direction is Lagrange's theorem, and the if direction is a general fact about *cyclic* groups we saw last semester.

- (5) Let k be a field of order p^d . Show that every element $a \in k$ is a zero for the polynomial $x^{p^d} - x \in \mathbb{F}_p[x]$.

Solution. Every non-zero element $\alpha \in k$ satisfies $\alpha^{p^d-1} - 1 = 0$ (why?). Multiplying by α gives $\alpha^{p^d} - \alpha = 0$. Clearly, $\alpha = 0$ also satisfies this equation.

- (6) Show that the polynomial $x^2 + 1$ is irreducible in $\mathbb{F}_p[x]$ if and only if $p \equiv 3 \pmod{4}$.

Solution. The polynomial $x^2 + 1$ is *reducible* in $\mathbb{F}_p[x]$ if and only if there exists $a \in \mathbb{F}_p$ such that $a^2 = -1$.

First suppose that $p = 2$. Then $1^2 = 1 = -1$. This tells us that we can exclude $p = 2$, and so assume that p is *odd*.

The factoring

$$(x^4 - 1) = (x^2 + 1)(x^2 - 1) = (x^2 + 1)(x - 1)(x + 1)$$

tells us that an element satisfying $a^2 = -1$ is exactly an element in $\mathbb{F}_p^\times$ of order 4 (why?). By the previous problem, $\mathbb{F}_p^\times$ contains such an element if and only if $p \equiv 1 \pmod{4}$. Therefore, $x^2 + 1$ is irreducible precisely when $p \equiv 3 \pmod{4}$.

- (7) Show that the polynomial $x^2 + x + 1$ is irreducible in $\mathbb{F}_p[x]$ if and only if $p \equiv 2 \pmod{3}$. Can you come up with a similar criterion for $x^2 - x + 1$?

Solution. Again, both these polynomials are quadratic and are reducible exactly when they admit zeros in $\mathbb{F}_p$. If $p = 3$, then 1 is a zero for $x^2 + x + 1$ and -1 is a zero for $x^2 - x + 1$. If $p = 2$, then neither polynomial has a zero in $\mathbb{F}_2$, so they are both irreducible in $\mathbb{F}_2[x]$.

We can now assume that p is an odd prime > 3 . The main observation involves the factorings

$$(x^3 - 1) = (x - 1)(x^2 + x + 1) ; (x^3 + 1) = (x + 1)(x^2 - x + 1).$$

This tells us that a zero in $\mathbb{F}_p[x]$ for $x^2 + x + 1$ is exactly an element of order 3 while one for $x^2 - x + 1$ is an element of order 6.

By problem 5, an element of order 3 can exist if and only if $p \equiv 1 \pmod{3}$, while an element of order 6 can exist if and only if $p \equiv 1 \pmod{6}$.

For the first polynomial, this leaves us with the option $p \equiv 2 \pmod{3}$ to ensure irreducibility, while for the second, it leaves us with $p \equiv 2, 3, 4, 5 \pmod{6}$. Of these options 2, 3, 4 are not possible (why?), and so we find the condition $p \equiv 5 \pmod{6}$. But actually this condition is equivalent to $p \equiv 2 \pmod{3}$ for odd primes (why?).

Actually, there's an easier way to see this equivalence between the two cases: If α is a zero for $x^2 + x + 1$, then $-\alpha$ is a zero for $x^2 - x + 1$, so one is irreducible if and only if the other is.